Grover lecture and lab

Instructor review guide for Rick Mukherjee
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Purpose: review the first of three planned sessions: **Grover; QFT/QPE; HHL/Shor**. The published rc1 ZIP contains the corrected first lesson. Dates, student preparation, assessment, and required native platforms still need confirmation.

Learning goals and preparation

Students should explain phase marking, interpret rightmost-bit/qubit-0 ordering, connect diffusion to reflection about the mean, compare exact probabilities with finite samples, and explain why extra iterations can reduce success.

Prerequisites: complex amplitudes, squared-magnitude probability, normalization, Hadamard gates, basis labels, and basic Python/Jupyter use. Complete installation and preflight before class. Use the code as a guided experiment; reading every assertion aloud would exceed the core time.

50-minute core

Time	Focus	Prediction or student action
0-5 min	Readiness and notation	Confirm the kernel; interpret 001 as index 1 and 011 as index 3. Identify qubits 1 and 0 in 011.
5-15 min	Phase oracle	Predict the sign change for target 001; verify that all uniform probabilities remain 1/8.
15-25 min	Diffusion and order	Relate D = 2|s><s| - I to reflection about the mean. Apply oracle first, then diffusion: Q = D O.
25-45 min	Single-target lab	Predict the four-state case; read the labeled circuit. Then check eight states/two iterations and sample 2,048 shots.
45-50 min	Exit discussion	Explain phase versus probability, operation order, retaining an unmeasured circuit, and overshooting.

75-minute version: add these 25 minutes

Time	Activity
50-65 min	Vary the two marked three-bit strings. Predict total success 1 and 1/2 per target after one iteration. Include the short invalid-input/duplicate-target activity within these 15 minutes.
65-75 min	Compare analytic and exact iteration curves, vary the marked count, and explain overshooting. Read the legend using the actual N and M.

Teaching procedure

Run all cells before teaching, then restart and clear outputs for the lesson. Ask for each prediction before execution. The labeled circuit identifies **Prepare**, **Oracle O**, and **Diffusion D**; the other figures show sampling and oscillatory success. Optional discussion does not mean skipped validation cells. A timed rehearsal is still needed.

Answer checkpoints and evidence

Review draft v0.2 | Mathematics, validation scope, and feedback for the next draft

Mathematical checkpoints

```
N = 2**n;  theta = asin(sqrt(M/N))
P_r = sin((2*r + 1)*theta)**2
O|x> = -|x> for marked x, and +|x> otherwise
D = 2|s><s| - I;  Q = D O;  initial state = |s>
```

Question or fixture	Answer
Why no probability change after the oracle alone?	The sign changes phase while leaving magnitude unchanged. Subsequent interference changes the probability distribution.
Diffusion update	Every complex amplitude becomes twice the mean minus its old value. The implementation fixes the exact D sign convention.
N=4, M=1; r=0, 1, 2	1/4, 1, 1/4. The third value demonstrates overshooting.
N=8, M=1; r=0, 1, 2	1/8, 25/32, 121/128 = 0.9453125.
N=8, M=2; r=1	Total success 1; each target has probability 1/2. This is unchanged when the distinct target pair changes.
Preparation and measurement	Prepare the uniform state once before repeating D O. Keep the source circuit unmeasured; sample a separate measured copy.

What is supported and what was checked

The canonical implementation supports **2-5 qubits**, explicit distinct binary targets, and nonnegative integer iteration counts. It rejects invalid types, empty/duplicate/all-marked sets, and malformed strings. It returns no measurements, classical bits, or ancillas. Enumerating targets is a teaching construction, not an end-to-end search benchmark or evidence of quantum advantage.

Executed in this review: 50 tests and four fresh-kernel notebooks passed on Linux CPython 3.12.14 with Qiskit 2.6.1 and Aer 0.17.2, reusing the pinned environment. The editable-target variations and dynamic legends are covered by five regressions. All nine Grover code cells executed and all three figures rendered.

Observed remotely: the PR checks and merged-commit classroom run passed. The latter used Ubuntu x64/CPython 3.12.15 and passed 50 tests/four notebooks. This review read those results; it did not trigger those runs.

Still unvalidated: native Windows/macOS, interactive notebook UI, hardware/GPU execution, and live classroom pacing. The local review reused packages; it was not a new installation. Remote success-path artifact upload was observed; deliberate remote failure-path testing was not performed.

Feedback requested from Rick

Schedule: actual dates and 50- or 75-minute meetings. **Preparation:** Python/Jupyter readiness, prior phase/amplitude coverage, devices and platforms. **Depth:** suitability of the guided lesson and later QFT/QPE; HHL/Shor sequence. **Assessment:** exploratory, submitted, or graded exercises and accommodation needs.

The Grover implementation is ready for QFT development to proceed while PDF integration, platform checks, and pedagogical feedback continue. Retain the next substantial technical review after QFT/QPE and before HHL/Shor.

Sources: Grover source commit 3894391; IBM Grover operator; IBM bit ordering.